

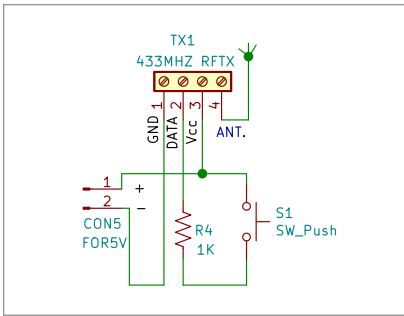


Fig. 2: Transmitter circuit

RF pulse. Additionally, pin 3 of the CD4017 is connected through a 100Ω resistor (R3) to the anode of LED1, with the cathode grounded. This LED1 indicates the relay's switching status.

Construction and testing

The actual-size PCB layout for the 433MHz RF relay switch is not provided here, so you can assemble the circuit on a general-purpose PCB for both transmitter and receiver. This allows you to build the circuit quickly at home.

You can alternatively use a standard 5cm × 7cm general-purpose Veroboard for assembly. Begin by placing a 16-pin IC socket on the board for the CD4017 IC (do not insert the IC at this stage). Then, mount the 433MHz RF receiver module onto the board. After that, place the remaining components, including resistors R1 (56k), R2 (330Ω), R3 (100Ω), LED1, transistor T1 (BC547), diode D1 (1N4001), and the 5V relay (RL1), onto the board and solder all connections carefully.

Ensure that the connections are neat and that there are no solder bridges. Avoid placing the CD4017 IC in the socket during soldering to prevent heat damage. After completing all components' soldering, you can insert the CD4017 IC into the socket. Ensure correct polarity for the diode and electrolytic capacitor, and confirm transistor orientation before soldering them.

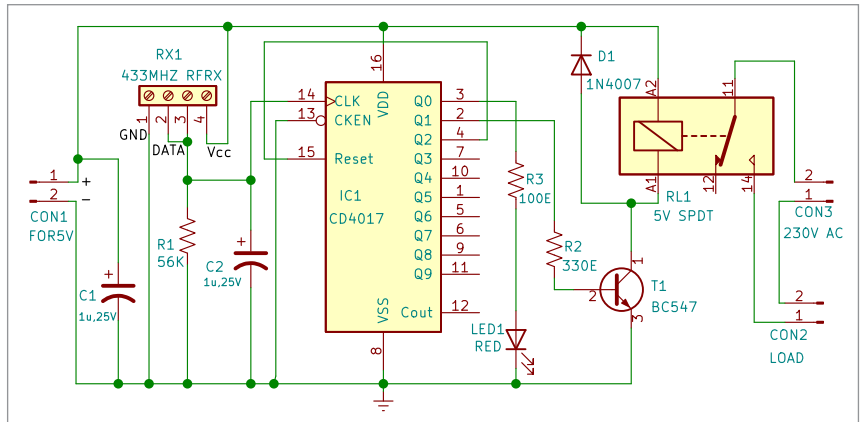


Fig. 3: Receiver circuit

For testing the circuit, you will need two separate +5V power supplies: one for the RF receiver and another for the transmitter. Connect the power supply to the respective power rails of your relay switch circuit. Also, power the 433MHz transmitter module with a 5V battery or supply. To test the load switching, connect a 230V AC lamp or any small appliance as a load to the relay contacts, ensuring you follow all safety precautions.

Now, bring the transmitter module near the receiver and press switch S1 on the transmitter side. Each time you press S1, LED1 on the receiver circuit will toggle; it will turn off when the relay is on, and turn on when the relay is off. This visual indicator confirms that the switching action is working correctly.

Performance and range

The operational range of this 433MHz RF relay switch is relatively short, typically limited to a few feet in indoor environments due to signal attenuation. For better range, ensure that a 7cm wire antenna is attached to the transmitter's antenna pad, and keep both transmitter and receiver modules free from metal enclosures or obstacles.

Caution. This circuit operates at 230V AC, which is a dangerous

PARTS LIST

Semiconductors:

- IC1 - CD4017 decade counter
- T1 - BC547 NPN transistor
- D1 - 1N4007 rectifier diode
- LED1 - 5mm red or green LED

Resistors (all 1/4-watt, ±5% carbon), unless stated otherwise:

- R1 - 56kΩ
- R2 - 330Ω
- R3 - 100Ω

Capacitors:

- C1 - 10μF, 25V electrolytic
- C2 - 1μF, 25V electrolytic

Miscellaneous:

- S1 - Push-to-on switch
- RL1 - 5V SPDT relay
- IC Socket - 16-pin socket for CD4017
- 7cm (2 No.) for RF transmitter and receiver)
- Connecting wires and soldering tools
- 5V DC power supply
- PCB - 5cm×7cm general-purpose Veroboard or custom PCB

and potentially lethal voltage. Always take proper electrical safety measures when testing or handling this circuit. Avoid touching live parts, and if you are unsure about working with high voltage, seek supervision or assistance from a qualified professional. Once tested and found working properly, you can safely integrate this circuit into your wireless control projects. **EFY**



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